EC 1745 Midterm Notes Name: _____

EXAM-DAY WORKFLOW (blank workbook, 60 min)

- Save as LAST_FIRST_EC1745MIDTERM. Name in A1. Ctrl+S after every part.
- One tab per part. Inputs typed once in labeled cells. Each answer is a formula that points at them, next to its label ("2.C.a").
- Rates as decimals (5% = 0.05). 1 bp = 0.0001. 100 bp = 0.01.
- Dates: **=DATE(2025,10,7)**. F4 adds $ (absolute ref). Ctrl+` shows formulas.
- PV/FV/PMT/RATE/NPER: cash in and cash out need opposite signs. Use **=-PV(...)** or enter pmt as negative.
- Self-checks: DuPont product = ROE. A = L + E. ΔCash from CF statement = ΔCash on BS. "Dumb" = "smart."

INCOME STATEMENT (build top-down)

Revenue (sales)	
– COGS	= **Gross profit**
– SG&A (– R&D)	= **EBITDA**
– Depreciation & amort.	= **EBIT** (operating income)
– Interest expense	= **Pretax income** (EBT)
– Taxes = t × EBT	= **Net income**
– Dividends	= Addition to retained earnings

EPS = NI / shares. Payout = Div / NI. Retention **b** = (NI – Div) / NI = 1 – payout.

BALANCE SHEET

Assets = Liabilities + Equity. Use it to plug one unknown (often cash).
CA = cash + AR + inventory. CL = AP + short-term debt. NWC = CA – CL. Operating NWC = AR + Inv – AP.
Net PP&E = gross PP&E – accum. depreciation. Equity(t) = Equity(t–1) + NI – Div + new shares – buybacks.

CASH FLOWS

CFO = NI + D&A – ΔAR – ΔInventory + ΔAP (= NI + D&A – ΔNWC)
CapEx = ΔNet PP&E + D&A (= ΔGross PP&E). **CFI** = –CapEx
CFF = ΔDebt + new stock – buybacks – dividends
ΔCash = CFO + CFI + CFF (check against the balance sheet)
FCF = EBIT(1 – t) + D&A – CapEx – ΔNWC
An asset ↑ uses cash. A liability ↑ is a source of cash.

RATIOS (end-of-year values unless told otherwise)

Gross margin	Gross profit / Sales
Operating margin	EBIT / Sales
Net profit margin	NI / Sales
ROE; ROA	NI / Book equity; NI / Total assets
Asset turnover	Sales / Total assets
Equity multiplier	Total assets / Book equity
DuPont ROE	(NI/Sales) × (Sales/TA) × (TA/Equity)
Sustain. growth	ROE × b [alt: ROE·b / (1 – ROE·b)]
Internal growth	ROA × b [alt: ROA·b / (1 – ROA·b)]
Current; quick	CA / CL; (CA – Inventory) / CL
Interest coverage	EBIT / Interest
D/E; D/capital	Debt / Equity; D / (D + E)
Days AR, Inv, AP	AR/(Sales/365); Inv or AP/(COGS/365)

If only D and E are given: TA = D + E. SGR: max growth with no new equity and constant D/E. IGR: max growth from retained earnings only.
DuPont says **why** ROE moved: margin (profitability), turnover (efficiency), multiplier (leverage; raises ROE and risk).

VALUATION MULTIPLES

Market cap (MV equity) = shares × price. **EV = MV equity + Debt – Excess cash**
MV equity = EV – Debt + Cash (EV = multiple × EBIT or EBITDA)
P/E = MV equity / NI = Price / EPS. Market-to-book = MV eq / BV eq

TIME VALUE OF MONEY

PV = FV / (1+r)^n FV = PV × (1+r)^n
Annuity (C at t = 1…N): PV = C/r × [1 – 1/(1+r)^N]
FV at N = C/r × [(1+r)^N – 1]
Perpetuity PV = C/r. **Growing perp.** PV = $C_1/(r - g)$, r > g
Growing annuity PV = $C_1/(r - g) \times [1 - ((1+g)/(1+r))^N]$
Annuity due (first payment today): PV = ordinary PV × (1+r).
Starts at t: value it at t – 1, then divide by (1+r)^(t–1).
Value at year T of any stream = PV × (1+r)^T. Compare deals at the **same date**. Highest PV wins. NPV rule: accept if NPV > 0.

EXCEL TVM ("SMART" COMMANDS)

=PV(r,n,pmt,[fv],[type])	Value today
=FV(r,n,pmt,[pv],[type])	Future value
=PMT(r,n,pv,[fv],[type])	Level payment (loan, coupon)
=RATE(n,pmt,pv,[fv])	Implied rate per period
=NPER(r,pmt,pv,[fv])	Number of periods
=NPV(r,CF1:CFn)	First CF is at t = 1! Use **=CF0+NPV(r,CF1:CFn)**
=IRR(CF0:CFn)	Implied rate, uneven CFs
=XNPV(r,CFs,dates)	**=XIRR(CFs,dates)** for dated CFs

type 0 = end of period (default), 1 = beginning (annuity due). Match r and n to the period: monthly uses APR/12 and 12 × years. Goal Seek: Data ▸ What-If ▸ Goal Seek.

"DUMB" TIMELINE METHOD (always works)

Row t = 0, 1, 2 … | row CF | row DF **=1/(1+$r)^t** | row PV = CF × DF. Total **=SUM(PV row)** = **=SUMPRODUCT(CFs,DFs)**.
Value at T: **=CF*(1+$r)^(T-t)**, then SUM.
Growing CFs: put **=C1*(1+g)^(t-1)** across the CF row. Implied rate: **=RATE(n,pmt,-PV)** for level CFs, or **=IRR(...)** with CF0 negative for uneven ones.

INTEREST RATES & LOANS (Ch. 5)

EAR = (1 + APR/k)^k − 1 **=EFFECT(APR,k)**
APR = k × [(1+EAR)^(1/k) − 1] **=NOMINAL(EAR,k)**
Periodic rate = APR/k. n-year rate = (1+EAR)^n − 1.
Continuous: EAR = e^APR − 1. Real: (1 + r_nom)/(1 + infl.) − 1
Loan payment **=PMT(APR/12,12*yrs,-loan)**. Balance after k payments = PV of the rest: **=PV(r,N-k,-pmt)**. Interest = r × prior balance; principal = pmt − interest.
Restructure a commitment: new deal = same PV at the same r. Solve for the unknown with PMT / RATE / NPER / Goal Seek.

BOND BASICS (Ch. 6)

Par = 100 (prices are % of par). Coupon/period = rate × 100 / f (f = 2 for semiannual).
YTM quoted as APR: per-period yield = y/f. **EAY = (1 + y/f)^f − 1**
P = Σ (c·100/f)/(1+y/f)^k + 100/(1+y/f)^N, N = f × years
Coupon > YTM → premium (P > 100). Coupon = YTM → par. Coupon < YTM → discount. **Issued at par ⇒ coupon = yield.**
Yield ↑ ⇒ price ↓. Price → 100 as maturity nears.
Zero: P = 100/(1+y/f)^N. YTM = f × [(100/P)^(1/N) − 1]
Corporate yield = Treasury yield + spread. More default risk means a wider spread and a lower price.
Ratings: BBB (Baa) and above = investment grade; below = high-yield (junk).
MV of debt = face × P/100. Face needed to raise $X = X / (P/100).
Spot rates: P = Σ CF_t/(1+y_t)^t. YTM is the one rate giving that price.

"DUMB" BOND COMMANDS (no dates)

Price	**=PV(y/2,2*T,-c*100/2,-100)**
Yield (APR)	**=2*RATE(2*T,c*100/2,-P,100)**
Coupon rate	**=2*PMT(y/2,2*T,-P,100)/100**
# periods	**=NPER(y/2,c*100/2,-P,100)**

Bond exchange: new bond must be worth the old price at the market yield. Solve for its coupon with PMT, or face = old value / (P_new/100).

"SMART" BOND COMMANDS (dates)

basis: 0 = 30/360 (corporates), 1 = actual/actual (Treasuries), 2 = act/360, 3 = act/365. frequency = 2.

=PRICE(s,m,c,y,100,2,b)	**CLEAN** price per 100
=YIELD(s,m,c,P,100,2,b)	YTM (APR) from clean price
=COUPPCD(s,m,2,b)	Previous coupon date
=COUPNCD(s,m,2,b)	Next coupon date
=COUPDAYBS(s,m,2,b)	Days since last coupon (A)
=COUPDAYS(s,m,2,b)	Days in period (E)
=COUPDAYSNC(s,m,2,b)	Days to next coupon (DSC)
=COUPNUM(s,m,2,b)	Coupons remaining
=DURATION(s,m,c,y,2,b)	Macaulay duration (yrs)
=MDURATION(s,m,c,y,2,b)	Modified duration

s = settlement, m = maturity, c = coupon rate, y = yield, P = clean price, b = basis. Format COUPPCD/COUPNCD cells as dates. **No real dates?** Use dummy dates exactly T years apart:
=DURATION(DATE(2000,1,1),DATE(2000+T,1,1),c,y,2).

CLEAN vs DIRTY PRICE

Accrued interest AI = (c·100/f) × A/E (E = 180 on 30/360)
Dirty (invoice) = Clean + AI. The dealer **quotes** the clean price; you **pay** the dirty price. PRICE() returns clean.
30/360 days = 30 × (months) + (day_2 − day_1). Example: 5/15 → 10/17 = 5×30 + 2 = 152.
AI does not depend on yield. New dirty price = PRICE(new y) + same AI.

DURATION

What: the PV-weighted average time (in years) until a bond's cash flows arrive.
Why: it measures interest-rate risk (price sensitivity to yields). Use it to compare bonds and to match asset and liability durations (immunization).
Macaulay D = Σ t × PV(CF_t) / P (t in years)
Modified MD = D / (1 + y/f)
%ΔP ≈ −MD × Δy $ΔP ≈ −MD × P × Δy
Example: MD = 7.7, yield +100 bp → price ≈ −7.7%.
From two prices: **D ≈ −[(P_2 − P_1)/P_1] / (y_2 − y_1)**
Zero-coupon D = its maturity. Longer maturity ↑ D. Higher coupon ↓ D. Higher yield ↓ D.
Convexity: actual price beats the estimate. It falls less when y ↑ and rises more when y ↓. The estimate works best for small Δy.

"DUMB" DURATION TABLE (then check with DURATION)

Columns: k | t (years) | CF | PV = CF/(1+y/f)^k | t × PV
Between coupon dates: periods = k − 1 + DSC/E; years = periods / f.
=SUM(PV) = (dirty) price. Macaulay D = **=SUM(t×PV)/SUM(PV)**. MD = D/(1 + y/f).

RECIPE: BENCHMARK + SPREAD QUESTION

1 **YIELD** of Treasury → 2 add spread → 3 **PRICE** of the corporate bond at that yield → 4 AI = coupon × COUPDAYBS/COUPDAYS → 5 dirty = clean + AI → 6 **DURATION**, **MDURATION** → 7 %ΔP ≈ −MD × Δy → 8 reprice to show actual (convexity).

SYNTAX CHECKS (past-exam values)

=PRICE(DATE(2024,10,17),DATE(2054,5,15),10%,12%,100,2,0) = 83.845; COUPDAYBS = 152; AI = 4.222; MDURATION = 7.804
=YIELD(DATE(2025,10,7),DATE(2035,8,15),4%,98,100,2,1) = 4.250%
=PV(3.5%,34,-2.75,-100) = 85.224. **=PV(10%,10,-30)** = 184.34